Alina Khan

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AI and Automation Engineer, CS undergraduate (Big Data & AI), experience in software and systems testing and in implementing AI agents and automation in an enterprise setting with real users every output verified and regression-gated, with self-healing recovery and runbooks the next engineer owns.

EXPERIENCE

Elchai Group | *AI Platform & Automation Engineering Intern* Jul 2026 – Sep 2026

- Deployed, optimized, and operated a production-grade AI platform and GenAI stack (Python, full-stack dashboard); implemented health checks, monitoring, self-healing cron, and automated recovery running unattended.
- Operated and tested Gate, a 3-model pipeline (GPT, Gemini, Claude) for classification, extraction, drafting. Added validation guards, REST retry logic, and an n8n self-healing workflow clearing a 200-task backlog without manual re-runs.
- Evaluated agent frameworks on latency, token cost, and observability; authored the decision matrix guiding agentic workflow adoption.
- Benchmarked emerging AI models on company use cases; built a Python/n8n multimodal eval pipeline with a regression gate. Shipped cron-driven automations and platform expansions (Slack PM agent, Stripe Subscription Tracker) that removed manual reporting steps.
- Built the company knowledge base (Obsidian, LLM wiki) and runbooks for agent token optimization; implemented and documented new Python tools and API workflows.

Delhi Private School, Sharjah | *Apprentice Intern — CHILD Department* 2023

- Supported departmental tasks and cross-team coordination

Trained on Odoo 19 ERP 2026

- Customized modules via CRUD operations, API integrations, and workflow automation.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C++
Agentic AI & Multi-Agent Systems	MCP, ReAct, LLM Orchestration, n8n, Hermes
RAG, KB & Evaluation	Obsidian/Graphify KB, Multimodal Model Benchmarking
Fine-Tuning & Post-Training	PyTorch, HuggingFace, PEFT, LoRA/QLoRA, Google Colab (A100/H100)
Production AI & Full-Stack	Inference Optimization, Quantization, vLLM, Docker, REST APIs, FastAPI, CI/CD, React, Next.js
Data & Databases	PySpark, Parquet, Pandas, NumPy, PostgreSQL, MongoDB, Redis
Tools	Git, GitHub, Vercel, Azure, Claude Code

PROJECTS

Aurum AI — The Gold Trading System | *Agentic AI Hackathon UOWD — 1st Place*

- Built and led a multi-agent gold-trading system using MCP for real-time market, macro, and news data. Built a ReAct agent council with confidence-weighted voting for trade signals, risk, sizing, and stop-losses, with persistent memory and a React paper-trading dashboard.

Arabic Sentiment Analysis via Transfer Learning | *CSCI316 Big Data Mining — UOWD 2026*

- Fine-tuned a 7B Arabic LLM (LoRA/PEFT, 4-bit quantization) on A100/H100 GPUs for code-switched Gulf Arabic-English sentiment analysis on 12.4K tweets.
- Developed the SDAS dialect-alignment metric and achieved 63.36% accuracy and 58.98% Macro-F1.
- Deployed the model through a Dockerized Flask API and interactive sentiment dashboard.

NYC Taxi Tip Prediction — Distributed ML Pipeline | *CSCI316 Big Data Mining — UOWD 2026*

- Built a Spark-based ML pipeline processing 11M+ NYC taxi trips using PySpark, Parquet, and Docker.
- Engineered temporal and spatial features and trained multiple regression models with custom cross-validation.
- Achieved best performance of $R^2 = 0.617$, RMSE = 2.375, and MAE = 1.258 on 1.4M test samples.

AI-Powered Innovation & Investment Platform | *Zayed University AI Hackathon 2025*

- Built an AI app using OpenAI embeddings and transformer models to match innovators with investors via cosine similarity on embeddings.
- Implemented real-time semantic search and idea discovery using HuggingFace Transformers with a full-stack web dashboard.

Autonomous Robot Navigation System | *Personal Project*

- Built a 3D Unity RL robot-learning environment with randomized obstacles and C# agent control, using a hybrid PPO + Fuzzy Logic + Behaviour Trees approach for emergent, adaptive obstacle-avoidance via large-scale training and domain randomization.

ACHIEVEMENTS

Presented Research Paper at ISDIA-2026 (International Conference) | *Brain Tumor Classification using CNNs*

Special Olympics Robotics — 3rd Place | *Programmed an EV3 robot for autonomous obstacle avoidance and object manipulation* 2019

Zayed University AI Hackathon — Selected Participant | *Chosen to build and pitch an AI platform under competition conditions* 2025

MCP & Agentic AI Hackathon — 1st Place | *UOWD, multi-agent gold-trading platform (MCP, ReAct)* 2026

EDUCATION

Bachelor of Computer Science (Big Data & Artificial Intelligence) | *University of Wollongong in Dubai* 2024–2027

Bachelor of Science in Biomedical Informatics (Transferred) | *University of Sharjah* 2023–2024

Delhi Private School, Sharjah 2023